



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

BT2102-1 – BIOPROCESS PRINCIPLES AND CALCULATIONS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Bar is a unit of pressure. Which of the following is equal to 1 bar?
A) 1.01325 atm
B) 101.325 kPa
C) 100 kPa
D) $1.01325 \times 10^5 \text{ N/m}^2$
2. How many grams of Carbon are present in 1 mole of glucose?
A) 72 grams
B) 180 grams
C) 1 gram
D) 12 grams
3. Molarity of sulfuric acid solution is 0.2 M. What is its normality?
A) 0.8 N
B) 0.4 N
C) 0.2 N
D) Data insufficient
4. "The distance between Nitte and Karkala is 10 Km."
Identify the error in the unit given in above statement.
A) "K" should be written as "k"
B) "m" should be written as "M"
C) "Km" should be written as "KM"
D) Between "10" and "Km" space should not be given
5. The process involving ideal gas with volume maintained constant is called as
A) Adiabatic
B) Isothermal
C) Isochoric
D) Isobaric
6. Compressibility factor of an ideal gas equals to
A) 0
B) >1
C) <1
D) 1
7. Which of these thermodynamic properties is a path function?
A) Temperature
B) Enthalpy
C) Internal energy
D) Heat supplied
8. In an adiabatic process, which of these is zero?
A) ΔV
B) Q
C) ΔU
D) W
9. For a process NOT involving any reaction under unsteady operation, which of the following term in mass balance equation is zero?
A) Rate of consumption
B) Rate of feed coming in
C) Rate of feed going out
D) Rate of accumulation
10. Drying operation can be termed as:
A) Vapor-Liquid-Solid operation
B) Vapor-Liquid operation
C) Solid-Liquid operation
D) Liquid-Liquid operation
11. The mass fraction of solids in the vapor stream in a drying process block diagram is
A) One
B) Between Zero and One
C) Zero
D) Data need to be provided / calculated
12. In a continuous mixing process, if two streams of different compositions are mixed, what principle is used to calculate the composition of the output stream?
A) Momentum balance
B) Mass Balance
C) Heat Balance
D) Energy Balance

13. In an open system at steady state, if there is no work interaction and no heat transfer, what happens to the energy balance?
- A) The total energy remains constant. B) The internal energy increases.
C) The kinetic energy decreases. D) The potential energy increases.
14. How much heat energy should be added to 10 kg/h fermentation medium to increase its temperature by 10°C? ($C_p = 4500 \text{ J/(kg.K)}$)
- A) 100 J/h B) 450 kJ/h
C) 450,000 kg/h D) 4500 J/h
15. Which of these relations between the specific heat at constant pressure (C_p) and specific heat at constant volume (C_v) is correct?
- A) $C_p < C_v$ B) $C_p = C_v$
C) $C_p > C_v$ D) $C_p = C_v + 0.5$
16. A cold fluid is heated using saturated steam at 121°C. The heat transfer takes place in an insulated chamber. Which of the following energy balance equations suits best for this case?
- A) $m_S \lambda = m_M C_p \Delta T$ B) $[m C_p \Delta T]_{\text{Steam}} = [m C_p \Delta T]_{\text{Cold}}$
C) $m_S \lambda = m_M C_p \Delta T + Q_{\text{Loss}}$ D) $[m C_p \Delta T]_{\text{Steam}} = [m C_p \Delta T]_{\text{Cold}} + Q_{\text{Loss}}$
17. Desired reaction: $A + B \rightarrow C$ Undesired reaction: $A + P \rightarrow Q$
Consider the above reactions in which, for every 4 moles of A fed, 2 moles of C and 0.5 moles of Q are produced. The selectivity in this case is
- A) 0.5 B) 4
C) 2 D) 0.25
18. Ultimate analysis of a fuel provides information about:
- A) The elemental composition, such as carbon, hydrogen, nitrogen, sulfur, and oxygen B) The moisture, ash, volatile matter, and fixed carbon content
C) The energy content of the fuel D) The combustion efficiency of the fuel
19. In the hydrolysis of ATP, ($\text{ATP} + \text{Water} \rightarrow \text{ADP} + \text{AMP}$) the change in water quantity is not measured. Because
- A) Water absorbs heat of reaction B) Water molecule acts as a catalyst
C) Water molecules are produced again D) Solvent Water is in large excess
20. Respiratory quotient for the following reaction is
- $$\text{C}_5\text{H}_5\text{COOH} + 5 \text{O}_2 + 0.5 \text{NH}_3 \rightarrow 0.5 [\text{C}_5\text{H}_7\text{NO}_2] + 2 \text{H}_2\text{O} + 4.5 \text{CO}_2$$
- A) 5 B) 1.25
C) 0.9 D) 0.5

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define normality and molarity with an example.	4	L*1	1	1
b)	How many kilograms of carbon are present in 64 kg of methane?	4	L3	1	2
c)	Calculate the heat and work in the following cases that involve ideal gas as working fluid.				
i.	At constant pressure of 1 bar, volume increases by 2 m ³ and temperature rises by 250 K.				
ii.	At constant volume of 1 m ³ , pressure increases by 2 bar and temperature rises by 250 K.				
iii.	At constant temperature of 500 K, the pressure doubles its initial value.				
iv.	An isolated system where initial pressure is 2 bar and the volume increase from 1 m ³ to 2 m ³ .				
Data: C_p for ideal gas = 29.32 J/(mol.K)		8	L4	2	1,2
2. a)	What are Intensive and Extensive properties in a thermodynamic process? Use a PV curve to explain an example for each case.	4	L2	2	1
b)	What are equations of states? Give an equation for Ideal gas and any two equations for gases that deviate from ideality.	4	L2	2	2

- c) A microbial growth medium of volume 500 mL is to be prepared. To prepare this medium 5 g of glucose, 1 g of K_2HPO_4 , 1 g of KH_2PO_4 , 0.2 g of $MgSO_4 \cdot 7H_2O$ were added. Determine the composition of medium in g/L and molar units.
Data: Atomic weight of K = 39, P = 31, O = 16, Mg = 24.3.
3. a) For a gas to be 'ideal,' what two main characteristics about its pushing force (pressure) and space it takes up (volume) need to be true?
b) Nitric acid and water forms maximum boiling azeotrope containing 62.2% water by mole. Find the % composition of the azeotrope by weight.
c) What is Carnot's cycle? Explain the cycle using a PV chart. Also, give the Carnot's Principles with relevant equations.

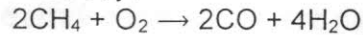
Unit – II

4. a) Caffeine is to be recovered from an aqueous solution using pure ethyl acetate (extract solvent) in a continuous, countercurrent, multiple-stage liquid-liquid extraction unit.
The available data as below:
• Feed: 250 kg/h with caffeine concentration of 0.10% w/w.
• Extract Solvent: 60 kg/h of pure ethyl acetate.
• Raffinate: 210 kg/h with caffeine concentration of 0.01% w/w.
Design the process by evaluating the remaining parameters and find out how much percent of caffeine can be recovered.
b) What is the specific heat capacity of a substance? Briefly describe established rule or formula used to approximate or estimate the specific heat for each of the following physical states:
i. Gases
ii. Solids
5. a) Draw the block diagram with suitable stream notations for the following unit processes. Also write the total and component balance equations.
i. Distillation operation with reflux
ii. Drying operation with bypass
b) A stream of nitrogen flowing at a rate of 10 kmol/h is heated from 303 K to 373 K. Calculate the heat that must be transferred to obtain this temperature increase. C_p is in kJ/kmol.K.
 C_p° for $N_2 = 29.5909 - 5.141 \times 10^{-3} T + 11.182 \times 10^{-6} T^2 - 4.968 \times 10^{-9} T^3$
6. a) Five hundred kilograms of a dilute waste acid containing 30% phosphoric acid is to be fortified to 50% phosphoric acid using concentrated phosphoric acid of strength 85%. How many kilograms of concentrated phosphoric acid is required for this process?
b) Describe the energy balance for a heat exchanger.
c) Draw the block diagram and material balance equations for a two stage process involving filtration operation followed by crystallization operation. The filtrate of stage 1 is the feed for stage 2. There is no water of hydration and crystallization takes place by cooling method.

Unit – III

7. a) What is Proximate analysis of fuels? List the components.
b) Calculate the standard heat of formation of benzoic acid using following data:
Std. heat of formation of $CO_2 = -393$ kJ/mol
Std. heat of formation of $H_2O = -285$ kJ/mol
Std. heat of combustion of benzoic acid ($C_7H_6O_2$) = -3226 kJ/mol

- c) A gaseous feed stream enters a continuous reactor containing 60 mole% Methane (CH_4) and 30 mole% Oxygen (O_2), with the remaining 10 mole% being Nitrogen (N_2) (inerts). In the reactor, 80% of the Methane reacts according to the following partial oxidation stoichiometry:



Calculate the composition of the product stream on a molar basis.

8 L3 5 2,3

8. a) What is Orsat analysis of flue gas? List out key points of analysis.
 b) Calculate the degrees of reduction for the following molecules: Hexadecane ($\text{C}_{16}\text{H}_{34}$), Benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$).
 c) Aerobic degradation of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) by a mixed culture of microorganisms ($\text{C}_5\text{H}_7\text{NO}_2$) can be represented by the following reaction:

4 L2 5 2,3

4 L3 5 2,3



Determine stoichiometric coefficients if $\text{RQ} = 0.9$.

8 L3 5 2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
